Grade 10 – Term 1
Comparing Decision Criteria: C8 Practice

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Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

1 Introduction

All problems use **profit payoffs**: larger values are better, and negative values represent losses. Alternatives A_i occupy rows; mutually exclusive states of nature S_j occupy columns. The payoff $P_{ay}(A_i, S_j)$ is the profit if A_i is chosen and S_j occurs. A decision is made before the state is known.

1.1 Four ways to compare the same alternatives

Maximax selects the alternative with the highest row maximum. It asks, "Which alternative offers the greatest possible profit?" **Maximin** selects the alternative with the highest row minimum. It asks, "Which alternative has the best worst-case profit?" Neither needs probabilities, even when probabilities are available.

Minimax Regret first compares alternatives within each state:

$$b(S_j) = \max_i P_{ay}(A_i, S_j), \qquad r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

Regret is the profit forgone compared with the best alternative for the state that actually occurs; it is not necessarily a cash loss. Select the alternative with the *smallest maximum regret*. This criterion needs the full payoff table, but no probabilities.

EMV/Bayes selects the highest Expected Monetary Value:

$$EMV(A_i) = \sum_{j=1}^{n} P(S_j)P_{ay}(A_i, S_j), \qquad P(S_j) \geq 0, \quad \sum_{j=1}^{n} P(S_j) = 1.$$

It requires a probability for every state as well as the payoffs. In these problems the state probabilities are the same for every alternative. Maximizing expected money expresses a risk-neutral orientation: it does not directly penalize uncertainty or protect against a large loss. An EMV is a weighted average, not a guaranteed profit or necessarily an attainable single outcome.

Criterion	Information and selection	Risk orientation	Main limitation
Maximax	Payoffs; highest row maximum	Optimistic; pursues upside	Ignores other outcomes and their likelihoods.
Maximin	Payoffs; highest row minimum	Conservative; protects the worst payoff	Ignores upside and how unlikely the worst state may be.
Minimax Regret	Full payoff table; smallest row maximum regret	Cautious about missed opportunities	Ignores likelihoods; depends on the alternatives included.
EMV/Bayes	Payoffs and valid probabilities; highest EMV	Risk-neutral about money	Sensitive to probability estimates; may recommend an unaffordable downside.

Criteria can disagree because they compare different summaries of the *same* information. Agreement does not mean that their reasoning is identical. Report all tied alternatives; do not invent a tie-breaking rule. Without probability information, EMV cannot be calculated unless an additional probability assumption is explicitly justified. Equal probabilities are not implied by uncertainty.

How to complete the numerical problems. The summary tables are already completed; do not recalculate row maxima, row minima, regrets, or EMVs. Identify the recommended alternative(s) by comparing the appropriate summary column. In the answer table, record the winning summary value and name the quantity it represents (for example, "Row minimum: 35"). Then justify and compare the recommendations using the supplied summaries and the original payoffs. When probabilities change, use the updated EMVs provided.



2 Comparing Decision Criteria

Problems:

2.1. Worked comparison: delivery expansion

2.2. Festival catering: changing probabilities

2.3. Retail plans: agreement and ties

2.1 Delivery Expansion

A delivery firm chooses its expansion plan. Quarterly profits are in thousand US dollars. Forecast probabilities are shown in the table.

Alternative	Strong (S_1)	Stable (S_2)	Weak (S_3)
$P(S_j)$	0.20	0.60	0.20
Large fleet (A_1)	100	20	-20
Flexible fleet (A_2)	75	50	10
Current fleet (A_3)	40	40	35
Contract partners (A_4)	60	65	20

Evaluating these payoffs and forecast probabilities under standard decision criteria yields the following summary statistics:

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	100	-20	55	28
A_2	75	10	25	47
A_3	40	35	60	39
A_4	65	20	40	55

Questions/Tasks



C8 Use the completed summary above to identify all four recommendations in the blank answer table. Compare their risk orientations, information requirements, and limitations using numerical evidence.

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Identify each criterion's recommendation using the supplied summary and record its deciding value.
2. Compare the recommendations. Explain each criterion's risk orientation and whether it needs probabilities or comparisons across alternatives within states.
3. Discuss a limitation of each criterion using the profits and summaries. Consider a firm unable to absorb a loss of 20.

Solution

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	A_1	Row maximum: 100
Maximin	A_3	Row minimum: 35
Minimax Regret	A_2	Maximum regret: 25
EMV/Bayes	A_4	EMV: 55

Maximax chooses the highest entry in the row-maximum column; Maximin chooses the highest row minimum; Minimax Regret chooses the smallest maximum regret; EMV/Bayes chooses the highest EMV. The stable state has the greatest probability and A_4 performs best in that state. The other three criteria do not use those probabilities. Maximax is optimistic about upside, Maximin protects the worst payoff, Minimax Regret limits missed opportunities, and EMV/Bayes is risk-neutral about money. The first two summaries use each alternative's payoffs; regret also compares alternatives within each state; EMV additionally requires valid probabilities.

There is no calculation error in this disagreement. Each recommendation answers a different question. A firm unable to absorb a loss of 20 may reject A_1 despite its upside. A_3 protects profit but can miss 60 in the strong state. A_2 limits missed opportunity rather than guaranteeing the highest minimum profit. A_4 maximizes expected profit but can still earn only 20.

More specifically, Maximax favors A_1's possible 100 while ignoring its minimum of -20 and EMV of 28. Maximin favors A_3's minimum of 35 while ignoring that its EMV of 39 is below A_4's 55. Minimax Regret favors A_2's maximum regret of 25, but its minimum profit is only 10, below A_3's 35; it also ignores the state probabilities and depends on which competing alternatives are included. EMV/Bayes favors A_4's 55 under this forecast, but 55 is not one of its possible payoffs (60, 65, and 20). Its recommendation depends on the probability estimates and does not guarantee the firm's required profit. If avoiding a loss of 20 is a binding constraint, exclude A_1 before choosing among the feasible plans; its unrestricted Maximax recommendation is infeasible.



2.2 Festival Catering: Changing Probabilities

A caterer chooses a large or small contract before knowing attendance. Profits are in thousand US dollars.

Alternative	High (S_1)	Low (S_2)
$P(S_j)$	0.30	0.70
Large contract (A_1)	80	10
Small contract (A_2)	45	35

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	80	10	25	31
A_2	45	35	35	38

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Use the supplied summary for probabilities $(0.30, 0.70)$ to identify all four recommendations and deciding values in the blank answer table. Report all ties.

2. For revised probabilities $(0.60, 0.40)$, the updated EMVs are 52 for A_1 and 41 for A_2. Identify the revised EMV/Bayes recommendation and deciding value and compare it with the original recommendation. Explain why the other three recommendations stay unchanged.
3. Compare all four criteria's risk orientations and information requirements. Discuss a limitation of each using numerical evidence. Compare the reasoning of Maximin and Minimax Regret. Does limiting maximum regret necessarily protect the lowest profit?
4. Recommend a criterion for a caterer prioritizing expected profit and another for a caterer prioritizing the worst payoff. Cite the relevant numerical evidence and one limitation of each choice.



2.3 Retail Plans: Agreement and Ties

A retailer compares three plans. Profits are in thousand CNY.

Alternative	Strong (S_1)	Normal (S_2)	Weak (S_3)
$P(S_j)$	0.25	0.50	0.25
Plan A_1	60	45	30
Plan A_2	50	40	20
Plan A_3	60	35	30

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	60	30	0	45
A_2	50	20	10	37.5
A_3	60	30	10	40

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Use the supplied summary to identify all four recommendations and deciding values in the blank answer table. Report every tie.
2. Compare A_1 with each other alternative in every state. Explain why A_1 is never worse, yet some criteria still allow a tie.
3. Explain why agreement on A_1 would not make the four criteria interchangeable. Identify which summary each criterion uses, explain its risk orientation and information requirements, and discuss a limitation of each using numerical evidence.



3 Contextual Practice

Problems:

3.1. Urban farm crop mix

3.2. Business reserve investment

3.3. Coastal ferry capacity

For each problem, interpret the completed summary to fill the blank answer table, including all ties. Do not recalculate the supplied summaries. Compare each criterion's risk orientation, information requirements, and limitations using numerical evidence. Explain why EMV/Bayes is unavailable when probabilities are not supplied.

3.1 Urban Farm Crop Mix

An urban farm chooses a crop mix for the next season. Profits are in thousand US dollars; the estimates come from a market forecast.

Alternative	Strong (S_1)	Stable (S_2)	Weak (S_3)
$P(S_j)$	0.50	0.30	0.20
Specialty crops (A_1)	80	40	-10
Balanced crops (A_2)	60	55	20
Staple crops (A_3)	40	45	35

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	80	-10	45	50
A_2	60	20	20	50.5
A_3	45	35	40	40.5

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Use the supplied summary to identify all four recommendations and deciding values in the blank answer table. Compare their risk orientations, information requirements, and limitations using numerical evidence. Explain why the largest possible profit, the best worst-case profit, and the smallest maximum regret need not belong to one crop mix.
2. The farm must retain at least 30 in profit in every state to meet its obligations. Use the supplied row minima to identify the feasible alternatives. Compare a recommendation under this constraint with the unrestricted EMV recommendation, citing their supplied minimum profits and EMVs.
3. A revised forecast gives probabilities $(0.20, 0.30, 0.50)$. The updated EMVs are 23 for A_1, 38.5 for A_2, and 39 for A_3. Identify the revised EMV/Bayes recommendation and deciding value. Explain what changes, what stays unchanged, and why reliable probability estimates matter.

3.2 Business Reserve Investment

A business invests part of its reserve for one year. The table gives *net gains*, in thousand US dollars, rather than final balances. All listed states are possible.

Alternative	Growth (S_1)	Stable (S_2)	Recession (S_3)
$P(S_j)$	0.60	0.25	0.15
Growth fund (A_1)	90	25	-30
Balanced fund (A_2)	60	40	5
Fixed deposit (A_3)	25	25	25

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	90	-30	55	55.75
A_2	60	5	30	46.75
A_3	25	25	65	25

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Use the supplied summary to identify all four recommendations and deciding values in the blank answer table. Compare their risk orientations, information requirements, and limitations using numerical evidence. Explain any agreement using the deciding values, rather than claiming that the criteria have the same risk orientation.
2. Compare the growth fund's EMV with its possible recession loss. Explain why expected gain does not guarantee that the reserve is protected.
3. The treasurer prioritizes the highest expected gain; the owner requires a nonnegative gain in every state. Use the supplied row minima to identify the owner's feasible alternatives. Give the treasurer a recommendation using EMV/Bayes. For the owner, compare the highest supplied EMV among feasible alternatives with the Maximin recommendation, and explain why the constraint alone does not select a unique plan. Justify each recommendation and discuss its limitation.

3.3 Coastal Ferry Capacity

A ferry operator selects a seasonal capacity plan. Profits are in thousand US dollars. Initially, no reliable probabilities are available.

Alternative	High (S_1)	Normal (S_2)	Low (S_3)
Add two ferries (A_1)	100	35	-25
Add one ferry (A_2)	75	55	10
Lease peak capacity (A_3)	55	48	30
Keep current fleet (A_4)	30	30	30

Alternative	Row Maximum	Row Minimum	Maximum Regret	EMV
A_1	100	-25	55	—
A_2	75	10	25	—
A_3	55	30	45	—
A_4	30	30	70	—

EMV cannot be determined because state probabilities are unavailable. Mark the EMV/Bayes recommendation and deciding value as unavailable.

Criterion	Recommended Alternative(s)	Deciding Value
Maximax	_____	_____
Maximin	_____	_____
Minimax Regret	_____	_____
EMV/Bayes	_____	_____

1. Use the supplied summary to identify the Maximax, Maximin, and Minimax Regret recommendations and deciding values in the blank answer table. Explain their disagreements and report any ties. State why EMV cannot yet be determined.
2. A later forecast supplies $P(S_1) = 0.20$, $P(S_2) = 0.30$, and $P(S_3) = 0.50$. These probabilities form a valid distribution. The updated EMVs are 18 for A_1, 36.5 for A_2, 40.4 for A_3, and 30 for A_4. Identify the EMV/Bayes recommendation and deciding value and compare it with the earlier recommendations. Explain each criterion's risk orientation, information requirements, and a limitation using numerical evidence.
3. The operator dislikes missing opportunities but has no trustworthy forecast. Which criterion fits that priority? Explain why it does not necessarily select the plan with the safest profit.



4 Mixed Practice

Problems:

4.1. Choosing a criterion

4.2. Evaluating decision claims

4.3. Decision committee report

4.1 Choosing a Criterion

For each situation, select the most appropriate of the four criteria *for the stated priority*. Explain its information requirements, risk orientation, and one limitation. If another criterion is feasible, explain why it serves a different priority.

1. A start-up has a complete payoff table but no reliable probabilities. Its founder wants the largest possible profit and accepts the downside.
2. A family business has reliable probabilities but prioritizes the highest worst-case profit, even if expected profit is lower.
3. A delivery manager has no reliable probabilities and wants to minimize the greatest profit forgone relative to the best plan in each state.
4. A retailer makes many comparable seasonal decisions, trusts the state probabilities, and prioritizes average monetary profit over time.
5. A museum knows its payoffs but has no attendance probabilities. Its director requests the plan with the highest EMV. Identify the missing information and explain why choosing equal probabilities would introduce an assumption.



4.2 Evaluating Decision Claims

Decide whether each statement is valid. Correct invalid statements and support your response with a numerical example from this activity where possible.

1. "Maximin and Minimax Regret must select the same alternative because both are cautious."
2. "If an alternative has an EMV of 55, it will earn 55 in the actual season."
3. "When probabilities change but payoffs stay fixed, all four recommendations must change."
4. "Regret of 25 means an actual monetary loss of 25."
5. "If Maximax and Bayes select the same alternative, Bayes is an optimistic criterion."
6. "A valid probability distribution is enough to make EMV the best criterion for every decision-maker."



4.3 Decision Committee Report

Return to the delivery expansion problem in Section 2.1. Four committee members prioritize, respectively, the largest possible profit, the best worst-case profit, the smallest maximum missed opportunity, and the highest expected profit.

1. Match each member to a criterion and an alternative. Give the numerical deciding value and explain which information the member uses or ignores.
2. Write a short recommendation to the firm that compares at least two competing plans. State your chosen decision priority, use numerical evidence, and acknowledge one limitation. Explain why another member can reasonably disagree.
3. Suppose the forecast is withdrawn. Identify which supplied summaries remain usable and which recommendation loses its information basis. Explain why this does not automatically make any one remaining criterion universally

best.

Self-check. A complete comparison identifies the available information, explains the decision-maker's priority, reports the recommended alternatives (including ties), and discusses the outcomes or uncertainty that each criterion leaves out.

